

3NF stands for Third Normal Form. A database is called in 3NF if it satisfies the following conditions:

- It is in second normal form.
- There is no transitive functional dependency.
- For example: $X \rightarrow Z$

Where:

$X \rightarrow Y$

Y does not $\rightarrow$ X

$Y \rightarrow Z$ so, $X \rightarrow Z$

44) What is BCNF?

BCNF stands for Boyce-Codd Normal Form. It is an advanced version of 3NF, so it is also referred to as 3.5NF. BCNF is stricter than 3NF.

A table complies with BCNF if it satisfies the following conditions:

- It is in 3NF.
 - For every functional dependency $X \rightarrow Y$, X should be the super key of the table. It merely means that X cannot be a non-prime attribute if Y is a prime attribute.
-

45) Explain ACID properties

ACID properties are some basic rules, which has to be satisfied by every transaction to preserve the integrity. These properties and rules are:

ATOMICITY: Atomicity is more generally known as 'all or nothing rule.' Which implies all are considered as one unit, and they either run to completion or not executed at all.

CONSISTENCY: This property refers to the uniformity of the data. Consistency implies that the database is consistent before and after the transaction.